External Backup GCP Design

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JIRA: CS-14029: External Backup GCP Design DONE

Requirements

NFR	Requirements	Acceptance Criteria
NFR-BR-003.03	Evidence shall be provided which shows where backups will be stored and the SLAs that have been agreed.	AWS Aurora - backups are taken and stored in the primary region, A **nightly** backup is taken and stored in the GCP UK region

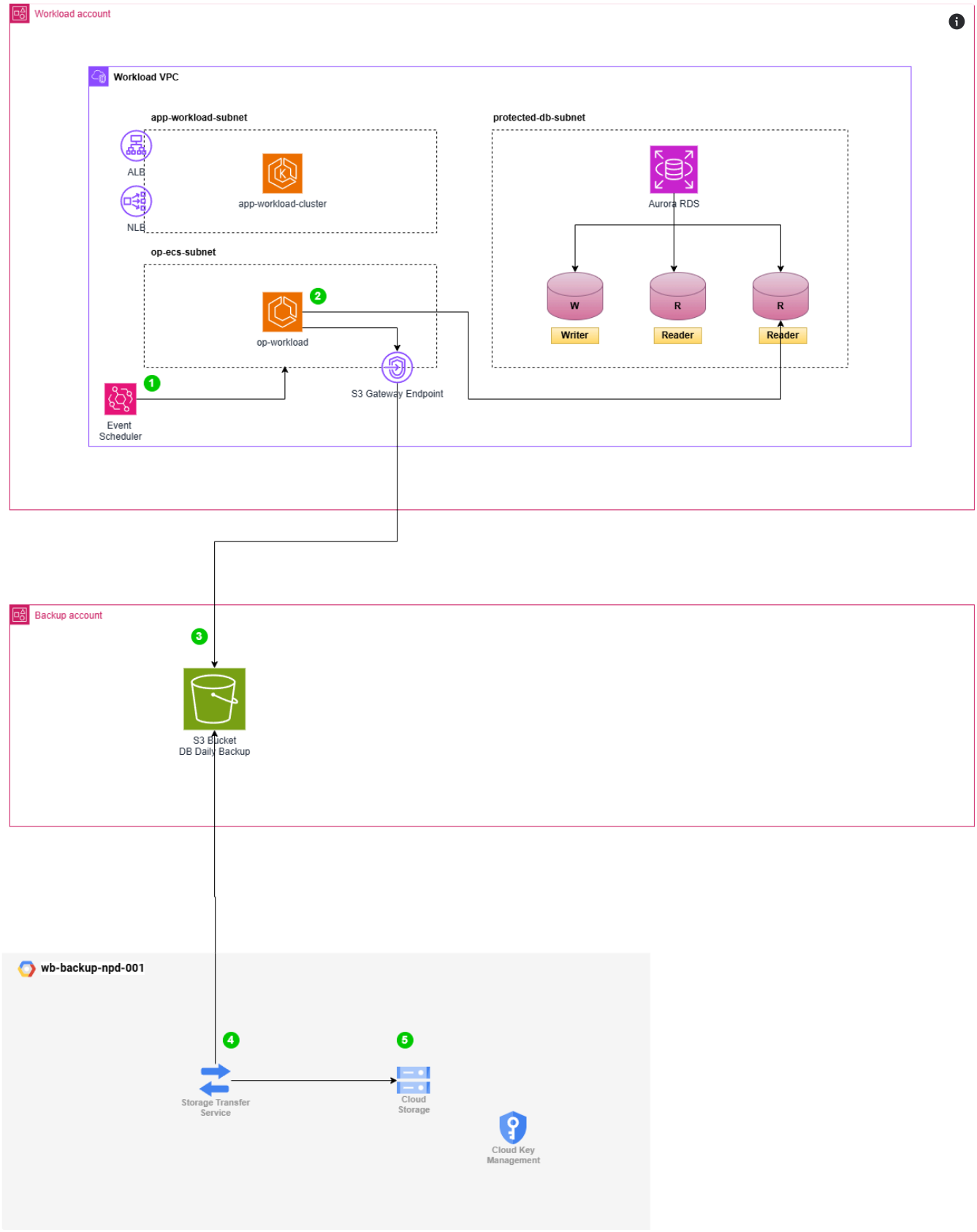
Objective

- Implement an automated nightly off-site backup solution to store Aurora Postgres backups in GCP

Provides cloud-level isolation and survivability in case AWS-native backups become unavailable or compromised.

High Level Architecture

The solution implements a secure, cross-cloud off-site backup strategy for Aurora PostgreSQL using logical backups (pg_dump) stored in AWS S3 and transferred to GCP via Storage Transfer Service (pull model).



Solution Components

Components	Purpose
Aurora PostgreSQL Read Replica	Aurora Database will be deployed in a DB-cluster mode with 1 Writer and 2 Read replica to ensure High availability. **Backups are taken from a read replica to avoid impacting the primary writer instance, ensuring production workloads are unaffected.**
ECS Fargate Task	Runs Postgres logical backup scripts in a containerized environment. • Offers a serverless, containerized environment for custom backup scripts. • Ephemeral, eliminating the need to manage EC2 instances. • Supports secure credential management with AWS Secrets Manager and network configuration with private access to database
S3 Bucket	Store the logical backup file in an immutable S3 bucket for GCP Transfer job consumption
GCP Storage Transfer Service	Automates transfer from AWS S3 to GCP Cloud Storage. GCP Storage Transfer Service is specifically designed to handle scheduled, automated data transfers between AWS S3 and GCP Cloud Storage, minimising operational overhead.
GCP Cloud Storage	Store the logical backup file in an immutable Cloud Storage bucket with retention and lifecycle policies according to

	compliance requirements

Aurora Backup Workflow

Step	Description
Scheduling	AWS EventBridge Scheduler triggers the backup process at a defined nightly window.
Backup Job execution	ECS Fargate task connects to the Aurora PostgreSQL read replica. Executes **pg_dump** to create a logical backup file and generate a checksum (SHA256) file to ensure backup file integrity. Uploads the backup file to an S3 bucket in a dedicated Backup AWS account
Off-site Transfer	GCP Storage Transfer Service runs on a schedule (scheduled appropriately after backup completes). Transfers new backup files from S3 to GCP Cloud Storage.
Storage & Retention	GCP Cloud Storage applies retention and lifecycle policies according to compliance requirements.

Implementation Considerations

Cloud	Considerations
AWS	• Deploy the ECS Fargate container in a dedicated subnet within the workload account to isolate operational tasks from core application workloads. • Configure the security group for Fargate and restrict Aurora cluster inbound connections to the Fargate security group. • Use a private gateway endpoint for Fargate to access the S3 bucket. • Encrypt the S3 bucket in the Backup account with customer-managed keys. • Enable Object Lock to store backups using a write-once-read-many (WORM) model to prevent objects from deletion or overwriting. • Define a cross-cloud IAM role with trust policy for GCP to authenticate S3 using `AssumeRoleWithWebIdentity`

	• Backup Scope: Aurora DB Backup and Cognito Backup
GCP	Create a **Transfer Job** in Scheduling mode with these parameters • **Source**: Amazon S3 • **AWS IAM Role ARN**: arn:aws:iam::937245949235:role/dev-gcp-transfer-poc • **Destination**: Google Cloud Storage • **Schedule**: <<StartTime>>, Frequency • **Network Option**: Google-managed private network • **Monitoring**: Enable Cloud Logging • **Notification**: Publish events on transfer status - Success, Failed, or Aborted. **Security**: • Transfer job network traffic using Google-managed private network with AWS S3 • Encrypt destination cloud storage buckets with a dedicated Cloud KMS key • Set retention policy and bucket to LOCKED mode to prevent deletion. • Dedicated GCP project for Parity off-site backup storage

Aurora Restore Workflow

> The proposed solution uses AWS DataSync to securely retrieve offsite PostgreSQL backups from GCP into Amazon S3, followed by a containerized restore process using ECS.

Step	Description
Initiate Data Sync Task	Start the AWS **Data Sync** task manually with Source location as GCP Cloud Storage authenticated using HMAC Access Key and Secret key. Configuring AWS DataSync transfers with Google Cloud Storage - AWS DataSync Initiate the transfer for pg dump backup file and checksum file
Prepare Target Aurora DB cluster	Provision a new Aurora DB Cluster and AWS Secrets for Database access.
Restore Job execution	ECS Fargate task download the pg dump backup file and checksum file Run checksum validation and initiate restore only if checksum passes ok.

	Executes **pg_restore** and restore the backup to the new targeted Aurora DB Cluster
Post Restore validation	Validate the Schema Objects, Key tables and run smoke test queries

Assumptions, Decisions and Dependencies

Type	Description
DEPENDENCY	Deloitte GCP team will create the necessary configurations in GCP for this backup workflow
DEPENDENCY	Enable SIEM monitoring in Google SecOps
ASSUMPTION	Backup retention policy for nightly offsite backup : 30 days
ASSUMPTION	The offsite backup copy in GCP will only be utilised in the event of a rare, widespread regional outage in Ireland. In such a scenario, Deloitte would need to re-provision the entire environment in a new region, with a RTO of 7 days and RPO (Recovery Point Objective): 24 hrs.
RISK	You can increase backup frequency to reduce RPO; however, as the database grows, if the logical dump export exceeds 4–6 hours, daily backups usually remain the practical limit. Increasing frequency may cause overlapping processes or require aggressive tuning.
DECISION	Leverage **GCP Storage Transfer Service** to facilitate the backup transfer from AWS to GCP using Google Managed Private network.
DECISION	Backup Scope: **Aurora DB Backup** and **Cognito Backup**
DECISION	The restore process will be one-time activity leveraging **AWS Data Sync** capability to transfer the off-site backup from GCP Cloud Storage to new AWS region S3 bucket. Configuring AWS DataSync transfers with Google Cloud Storage - AWS DataSync

References

- **GCP Storage Transfer Service POC :** GCP - Storage Transfer Service Config - Converge Banking - Confluence
- **Cognito backup and Restore:** Implementation Runbook : Cognito Backup & Restore
- **Backup File Integrity check** : How to Verify Backup File Integrity in Linux | Step-by-Step Guide